

Experiment 5 :

Pogramm:-

```
#include<stdio.h>
```

```
void firstFit(int blockSize[], int m, int processSize[], int n)
{
    int allocation[n];

    for(int i = 0; i < n; i++)
        allocation[i] = -1;

    for(int i = 0; i < n; i++)
    {
        for(int j = 0; j < m; j++)
        {
            if(blockSize[j] >= processSize[i])
            {
                allocation[i] = j;
                blockSize[j] -= processSize[i];
                break;
            }
        }
    }

    printf("\nFIRST FIT\nProcess No.\tProcess Size\tBlock No.\n");
    for(int i = 0; i < n; i++)
    {
        if(allocation[i] != -1)
            printf("%d\t%d\t%d\n", i+1, processSize[i], allocation[i]+1);
        else
            printf("%d\t%d\t\tNot Allocated\n", i+1, processSize[i]);
    }
}
```

```
void bestFit(int blockSize[], int m, int processSize[], int n)
{
    int allocation[n];

    for(int i = 0; i < n; i++)
        allocation[i] = -1;

    for(int i = 0; i < n; i++)
    {
        int bestIdx = -1;
```

```

    for(int j = 0; j < m; j++)
    {
        if(blockSize[j] >= processSize[i])
        {
            if(bestIdx == -1 || blockSize[j] < blockSize[bestIdx])
                bestIdx = j;
        }
    }

    if(bestIdx != -1)
    {
        allocation[i] = bestIdx;
        blockSize[bestIdx] -= processSize[i];
    }
}

printf("\nBEST FIT\nProcess No.\tProcess Size\tBlock No.\n");
for(int i = 0; i < n; i++)
{
    if(allocation[i] != -1)
        printf("%d\t\t%d\t\t%d\n", i+1, processSize[i], allocation[i]+1);
    else
        printf("%d\t\t%d\t\t\tNot Allocated\n", i+1, processSize[i]);
}
}

int main()
{
    int blockSize[] = {100, 500, 200, 300, 600};
    int processSize[] = {212, 417, 112, 426};

    int m = 5;
    int n = 4;

    int block1[5], block2[5];

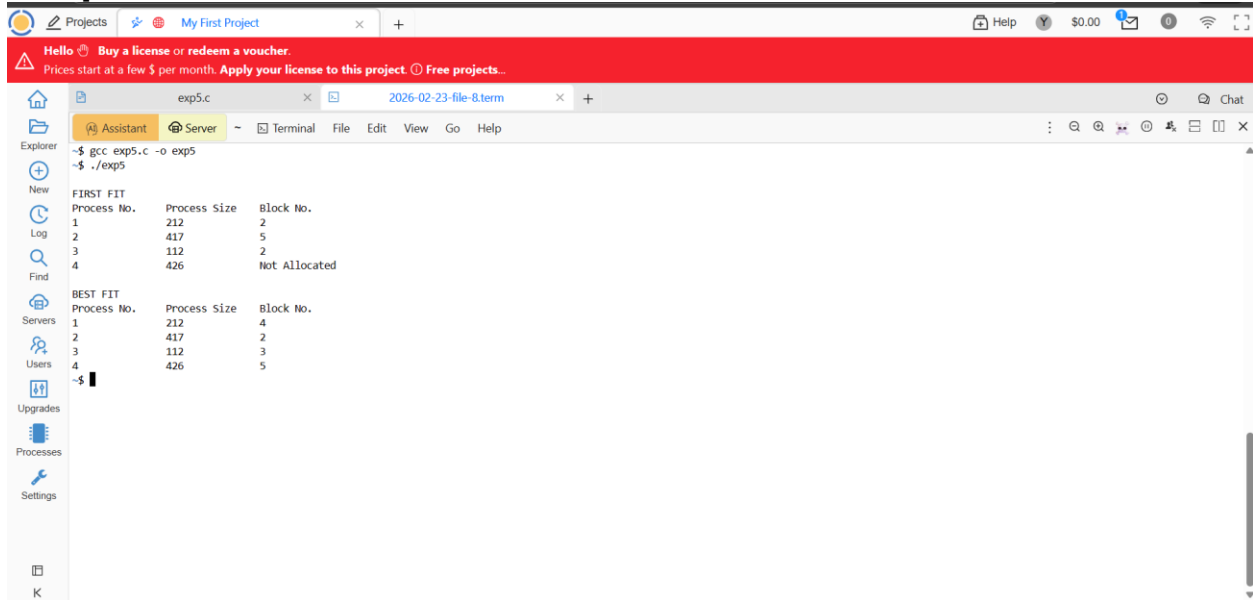
    for(int i = 0; i < m; i++)
    {
        block1[i] = blockSize[i];
        block2[i] = blockSize[i];
    }

    firstFit(block1, m, processSize, n);
    bestFit(block2, m, processSize, n);
}

```

```
    return 0;
}
```

Output:-



```
exp5.c 2026-02-23-file-8.term
Assistant Server ~ Terminal File Edit View Go Help
~$ gcc exp5.c -o exp5
~$ ./exp5

FIRST FIT
Process No. Process Size Block No.
1 212 2
2 417 5
3 112 2
4 426 Not Allocated

BEST FIT
Process No. Process Size Block No.
1 212 4
2 417 2
3 112 3
4 426 5
~$
```